Project 1

In this project, you will design and implement web-based interactive techniques to show related information between a table and text.

Text and table are frequently used together to help people make sense of data. For example, several paragraphs may describe GDP growth across the world over the past 50 years, while a table summarizes the same growth figures. Although the text and table are related (e.g., specific numbers mentioned in the text also appear in the table), they are visually separate. As a result, readers must constantly switch their attention back and forth between the two.

To address this problem, you will design and create a solution that helps users easily connect related information between text and table. Your specific tasks are:

1) Find or create a dataset that contains both textual information and a data table.

- The textual information must be at least 1000 words long and organized into 6-8 paragraphs.
- The data table must have at least 6 columns and 20 rows.
- At least 30 cell values from the table must be mentioned in the text. This requirement ensures that the text and table are meaningfully related.

2) Display both the text and the table on a single webpage.

3) Design and implement interactive techniques that help readers discover and navigate the connections between the text and the table. For example:

- When a user encounters numbers in the text, they should be able to quickly locate the corresponding cells in the table.
- When a user examines cells in the table, they should be able to find where (if at all) those values are mentioned in the text.

You must implement your design using web technologies (HTML, CSS, and JavaScript). You are encouraged to explore and use web frameworks. If your solution builds on existing examples, libraries, or work created by others, you **must** clearly acknowledge those sources and provide proper references in your submission. Follow good software-engineering practices (e.g., separate content, style, and behavior).

Write a short document that describes: 1) your design decisions corresponding to the three tasks above, 2) how to deploy and run your code, and 3) how does each group member contribute to the submission. The document should be prepared in Microsoft Word using Times New Roman, 12-point font, single line spacing, and must not exceed 4 pages.

You are required to complete this project in a group of 2-4 students. One group member will submit the work and must ensure that the names of all group members are included in the submission. Your submission should include both the complete source code and the required design document.